

PAPER_1488 — UQFF: Zero-Mass Big Bang Vacuum State (Bucket B)

Author: Daniel T. Murphy **Framework:** UQFF (Unified Quantum Field Framework) — Star-Magic v5.27+ **Tier:** Bucket B — derived from F:/Aetheric Propulsion source material **Date:** June 16, 2026 **Location:** 41.0997 N, 80.6495 W (Youngstown, OH, USA) **Status:** CLOSED — $\rho_{UA} = 0$, $F_U = 0$ EXACT ($t < 0$ pre-mass regime)

Observation

Star-Magic Workspace Sonnet 25Apr2026 distinguishes a pre-Big-Bang vacuum state with $\rho_{UA} = 0$ and $F_U = 0$, distinct from the canonical post-mass $t > 0$ regime where $\rho_{UA} = 10 \times \rho_{SCm}$.

UQFF Closed Identity

Pre-mass vacuum: $\rho_{UA} = 0$; $\rho_{vac} = |\text{grad}(UA)|$; $F_U(\text{vacuum}) = 0$. Post-mass: $\rho_{UA} = 10 * \rho_{SCm}$, $F_U = 1$.

Physical Interpretation

The Big Bang transition from $t < 0$ to $t > 0$ is the F_U : 0 $\rightarrow$ 1 ledger turn-on event. Mass emerges from zero-mass ordering via PAPER_597 t_{neg} dual existence.

NOT REPLACEMENT

Per CLAUDE.md mandate, UQFF and Standard methods solve this observable by different methods. Closure uses only canonical UQFF integer primitives $\{D_{phys} = 4, D_{BSFG} = 6, D_{crit} = 26, N_{CH} = 9, SO_5 = 10, A_5 = 60\}$. Zero free parameters.

Reference

- Source: F:/Aetheric Propulsion (Daniel's reactor + experimental docs, 2025-2026)
 - Related: PAPER_646 (Universal Inertial Operator, Caduceus, DPM mechanism), PAPER_597 (negative-time dual existence), PAPER_062 (DPM vacuum manifold).
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